

Digital Electronics (II SEM B.Tech. CSE)

Couse Code: NECE102



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Digital Electronics: Course Goals and Key Takeaways



Course Objective

To develop an understanding of digital electronics and its application which will be further useful to in the advanced courses of Electronics and Industrial Engineering.

Learning Outcomes

Upon successful completion of this course, students will:

- Gain information about the basic knowledge of digital electronics to design various types of digital circuits and systems.
- Learn how to analyse the performance and working of digital circuits.
- Develop their knowledge of digital electronics for industrial and robotic applications.

Digital Electronics: Key Topics at a Glance



Unit No.	Topics to be Covered	Lecture Hours	Learning Outcome
1.	Number System, Logic Gates, Boolean Algebra, NAND and NOR Implementation of Logic Functions, Determination of Boolean Functions from Digital Circuits.	08	Acquire an understanding of number system, basic logic gates and their applications to develop digital circuits and systems.
2.	Standard Representations of Logical Functions, SOP and POS form, Simplification of Logical Functions using K-map.	08	Acquire the knowledge to represent the logic functions in standard form and its minimization.
3.	Combinational Logic Circuit: Adder, Subtractor, Magnitude Comparator, Code Converters etc. MSI Circuits: Multiplexers, Demultiplexers, Decoders, Encoders etc.	08	Understand the design and applications of combinational logic circuits.
4.	Sequential Circuits: Latches and Flip-flops, Counters, Shift Registers.	10	Acquire the ability to design and perform the analysis of different sequential circuits.
5.	Analysis of Sequential circuits: State tables and state diagrams, Timing Circuits	04	Acquire an understanding on the analysis of sequential circuits.
6.	Introduction to logic families, semiconductor memories, ROM, PAL, PLA and Microprocessor	04	Develop knowledge about the various techniques for IC designing for digital circuits.

Textbook:

1. Floyd T.L., Digital Fundamentals, 10/2, Pearson Education, 2011.

Reference Books:

1. Digital Logic and Computer Design by M. Morris Mano, Prentice –Hall, 2015.
2. Fundamental of Digital Electronics by A. Anand Kumar, 4/e Prentice Hall, 2016.

Digital Electronics: Lecture Plan



Sl. No.	Topics	No of classes (tentative)
1.	Introduction to Number Systems and Conversion Techniques	8
2.	Continue.....	
3.	1's and 2's complement	
4.	Logic Gates (AND, OR, NOT, NAND, NOR, XOR, XNOR)	
5.	Boolean Algebra: Laws and Theorems	
6.	Boolean Function Implementation Using NAND/NOR Gates	
7.	Determination of Boolean Functions from Digital Circuits	
8.	Continue.....	
9.	SOP Standard Forms of Logical Functions	8
10.	POS Standard Forms of Logical Functions	
11.	Simplification of Logical Functions Using Boolean Algebra	
12.	Introduction to Karnaugh Maps (K-Maps)	
13.	K-Map Simplification (2-variable Functions)	
14.	K-Map Simplification (3-variable Functions)	
15.	K-Map Simplification (4-variable Functions)	
16.	Don't Care Conditions in K-Map Simplification	

Digital Electronics: Lecture Plan



17.	Introduction to Combinational Logic Circuits	8
18.	Design of Adders (Half Adder, Full Adder)	
19.	Design of Subtractors (Half Subtractor, Full Subtractor)	
20.	Magnitude Comparator Design	
21.	Code Converters (Binary to Gray, Gray to Binary, BCD to Excess-3)	
22.	Multiplexers: Design and Applications	
23.	Demultiplexers: Design and Applications	
24.	Decoders and Encoders	
25.	Introduction to Sequential Circuits	10
26.	Latches: SR Latch, D Latch	
27.	Flip-Flops: SR, JK Flip-Flops	
28.	Flip-Flops: D and T Flip-Flops	
29.	Asynchronous Counters: Design and Operation	
30.	Continue.....	
31.	Synchronous Counters: Design and Operation	
32.	Continue.....	
33.	Shift Registers: Types and Applications	
34.	Continue.....	

Digital Electronics: Weightage



Weightage of Different Components

1.	Quiz/ Assignment	10 Marks
2.	Mid Semester Examination	30 Marks
3.	Quiz/ Assignment	10 Marks
4.	End Semester Examination	50 Marks
Total Marks = 100		

Introduction to Digital Electronics



What is Signal..??

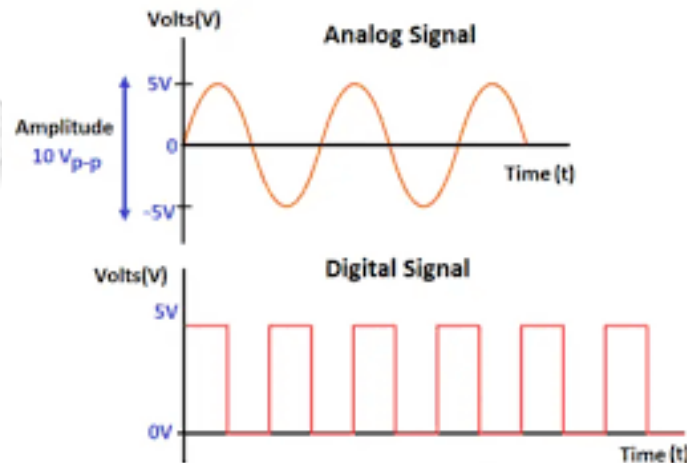
A signal is a function, that represents a variation of the physical quantity with respect to any parameter.

Independent quantity (generally time or distance)

- In electrical and electronics, usually the signal is the variation of the electrical quantity generally current or voltage over time.
- If the current or the voltage remains the same for different time then it is not a signal, i.e., if $dI=0$ Then it is direct current.

What is Analog Signal..??

- An analog signal is continuous, meaning it represents data in a continuous flow. It can take any value within a certain range.
- It is typically represented as a smooth, continuous wave, often a sine wave.



What is Digital Signal..??

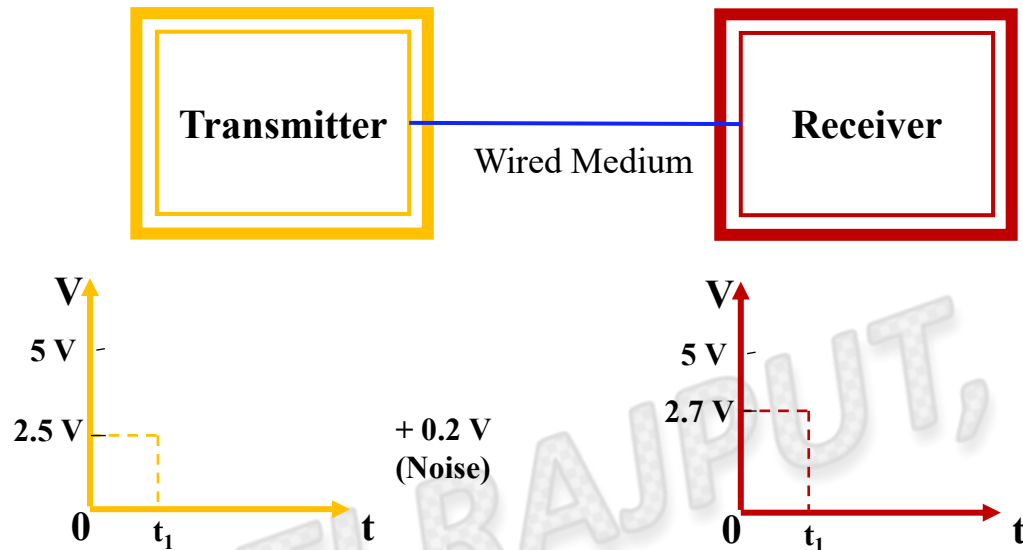
- A digital signal is discrete, meaning it represents data in the form of binary values (0s and 1s). These values are distinct and do not vary continuously.
- Digital signals are represented as a series of pulses, where each pulse corresponds to a binary state (high or low).

Introduction to Digital Electronics



Need of Digital Signal

- ✓ All real life signals are analog.
- ✓ Digital signal is used in communication process to minimize the effect of noise (unwanted signal).



Example: Let us say that we have the analog signal : $[0 \text{ V}-5 \text{ V}]$; and we need to transmit 2.5 V .

Note: This 2.7 V is considered to be correct because we are sending the analog signal and in case of analog signal, the voltage can take any value within the given limit.

So, there is effect of noise in case of analog signal.

Introduction to Digital Electronics



Electronic Circuits

Digital Circuits
(quantities with discrete values)

Analog Circuits
(quantities with continuous values)

Advantages of Digital Electronics

- **Digital electronics** is a branch of electronics which deals with digital format of data and codes.
- Digital stands for digit, digital electronics has two conditions **0 (low logic)** and **1 (high logic)**.
- Digital electronics system use a digital signal that are composed of mathematical features to work.
- “1” as true and “0” as false are called bit and the group of bits are named Byte.

- Computer-controlled digital systems can be controlled by software, allowing new functions to be added without changing hardware.
- Information storage can be easier in digital systems than in analog ones.
- The noise-immunity of digital systems permits data to be stored and retrieved without noise.
- Digital system are easier to design and more precise representation of signal can be obtained by using more binary digits to represent it.
- More digital circuitry can be fabricated on IC chips.
- Error management method can be inserted into the signal path. To detect signal errors, and then either correct the errors, or at least ask for a new copy of data.

Introduction to Number System



NUMBER SYSTEM

It defines a set of values used to represent the quantity.

System	Base (r)	Symbols 0 to (r – 1)	Used by Humans ?	Used in Computers ?
Decimal	10	0, 1, 2, 3, 4, 5, 6, 7, 8, 9 → (10 different digits)	Yes	No
Binary	2	0, 1 → (2 different bits)	No	Yes
Octal	8	0, 1, 2, 3, 4, 5, 6, 7 → (8 different digits)	No	No
Hexadecimal	16	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F → (16 different digits) 10 = A; 11 = B, 12 = C, 13 = D, 14 = E, 15 = F	No	No

Example: $(7392)_{10}$ → Decimal Number System is the most popular and common number system used in our daily life (distance measurement, weight measurement, money counting etc..)

Base:

- Base of the number system is also called as Radix (r).
- It will tell us how many distinct digits we have in a number system.

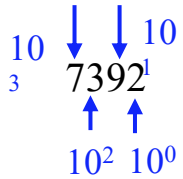
Introduction to Number System



Weighted and Unweighted Codes

Example: $(7392)_{10}$

$$7392 = 7000 + 300 + 90 + 2 = 7 \times 10^3 + 3 \times 10^2 + 9 \times 10^1 + 2 \times 10^0$$

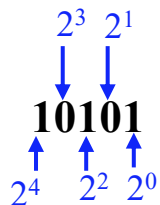


The blue color represents the weight of the position. Whereas 7, 3, 9, 2 are referred as the coefficients.

- ✓ The number systems with weights on the position are referred to as the **Weighted Number System**.
Example: Decimal, Binary, Octal, Binary Coded Decimal (BCD Codes) etc..
- ✓ The number systems with no weights on the position we call them as the **Unweighted Number System**.
Example: Gray Code, Excess-3 Code etc..

Binary Number System

- ✓ A computer cannot operate on the decimal number system → Thus do their operation on the binary logic.
- ✓ Base $(r) = 2$; Number of distinct bits is 0 to $(r-1)$ i.e., 0 and 1 (**Binary Digits are called Bits**)

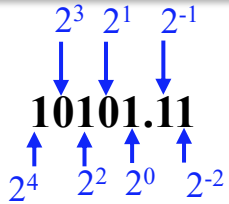


$$(10101)_2 = 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 16 + 0 + 4 + 0 + 1 = (21)_{10}$$

Binary Number

Decimal Equivalent

Binary Number System



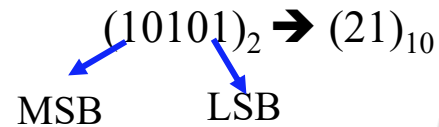
$$(10101.11)_2 = 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 + 1 \times 2^{-1} + 1 \times 2^{-2} = 16 + 0 + 4 + 0 + 1 + 0.5 + 0.25 = (21.75)_{10}$$

Binary Number

Decimal Equivalent

Binary Point

Most Significant Bit (MSB) and Least Significant Bit (LSB)



Why it is termed as MSB or LSB...??

Example: $(10101)_2$ **Replace LSB** from 1 to 0 and see the decimal equivalent.

$$(10100)_2 = 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0 = 16 + 0 + 4 + 0 + 0 = (20)_{10} \rightarrow \text{Not much difference}$$

$(10101)_2$ **Replace MSB** from 1 to 0 and see the decimal equivalent.

$$(00101)_2 = 0 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 0 + 0 + 4 + 0 + 1 = (5)_{10} \rightarrow \text{Much difference}$$

$b_4 \ b_3 \ b_2 \ b_1 \ b_0$

Bits are the smallest unit of the data.

1 Nibble = 4 Bits

1 Byte = 8 Bits

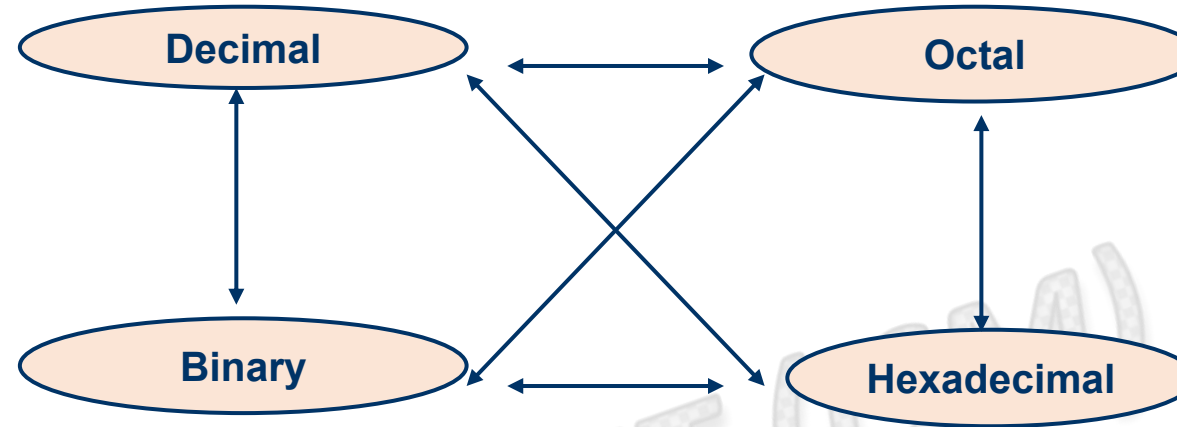
1 Word = 16 Bits = 2 Bytes

1 Double Word = 32 Bits = 4 Bytes

Conversion of Number System



Conversion among Bases



Binary to Decimal Conversion

- Multiply each bit by 2^n , where n is the “weight” of the bit.
- The weight is the position of the bit, starting from 0 on the right.
- Add the results

a.) $(11111)_2 = 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = 16+8+4+2+1 = (31)_{10}$

b.) $(1100.1011)_2 = 8+4+0+0+0.5+0+0.125+0.0625 = (12.6875)_{10}$

Conversion of Number System



Decimal to Binary Conversion

- To convert decimal to any other base 'r', divide integer by r and multiply fractional part by 'r'.
- For Decimal to Binary Conversion → Divide by two, keep track of the remainder
- First remainder is bit 0 (LSB, least-significant bit)
- Second remainder is bit 1

$$25_{10} = ?_2$$

Divisor	2	25	Divider
	2	12	1 ← First remainder
	2	6	0 ← Second Remainder
	2	3	0 ← Third Remainder
	2	1	1 ← Fourth Remainder
		0	1 ← Fifth Reaminder

Read Up

Binary Number = 11001

Convert 0.3125_{10} to binary

	Result Digit
$.3125 \times 2 = 0.625$	0 (MSB)
$.625 \times 2 = 1.25$	1
$.25 \times 2 = 0.50$	0
$.5 \times 2 = 1.0$	1 (LSB)

$$0.3125_{10} = .0101_2$$

Circuit Globe

Conversion of Number System



First Sixteen Decimal, Binary, Octal, and Hexadecimal Numbers

Decimal	Binary	Octal	Hexadecimal
0	0	0	0
1	1	1	1
2	10	2	2
3	11	3	3
4	100	4	4
5	101	5	5
6	110	6	6
7	111	7	7
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

Problem 1: Convert $(11011.101)_2$ to Decimal.
27.625

Problem 2: Convert $(1001011)_2$ to Decimal.
75

Problem 3: Convert $(52)_{10}$ to Binary.
110100

Problem 4 : Convert $(105.15)_{10}$ to Binary.
1101001.001001

Conversion of Number System



Decimal to Octal Conversion

- Divide integer by 8 and multiply fractional part by 8.
- Convert $(112)_{10} \rightarrow (160)_8$
- Convert $(25.625)_{10} \rightarrow (31.50)_8$
- Convert $(2048)_{10} \rightarrow$
- Convert $(82.250)_{10} \rightarrow$
- 4000
- 122.2

Decimal to Hexadecimal Conversion

Decimal Integer

- Divide integer by 16 till the quotient is zero.
- The last remainder is the MSB.
- Remainder read from bottom to top $\rightarrow$ Equivalent Hexadecimal Number

Decimal Fraction

- Successively multiply the given decimal fraction and subsequent decimal fractions by 16, till the product is zero or till the required accuracy is obtained.
- Collect all the integers to the left of decimal point
- The first integer is MSB and the integers read from top to bottom give the hexadecimal fraction.

Conversion of Number System



Decimal to Hexadecimal Conversion

Problem 1 : Convert $(2598.675)_{10}$ to Hexadecimal.

Successive division

16	2598
16	162
16	10
	0

<i>Remainder</i>	
<i>Decimal</i>	<i>Hex</i>
6	↑ 6
2	2
10	A

0.675×16
 0.800×16
 0.800×16
 0.800×16

| 10.8
| 12.8
| 12.8
↓ 12.8

Problem 2: Convert $(294.125)_{10}$ to Hexadecimal.
 $(126.2)_{16}$

Conversion of Number System



Octal to Binary Conversion

To convert a given octal number to a binary, just replace each octal digit by 3-bit binary equivalent.

Problem 1 : Convert $(367.52)_8$ to Binary

Answer: $(011110111.101010)_2$

Problem 2 : Convert $(273.15)_8$ to Binary

Answer: $(010111011.001101)_2$

Binary to Octal Conversion

To convert a binary number to an octal number, starting from the binary point make groups of 3 bits each, on either side of the binary point, and replace each 3-bit binary group by the equivalent octal digit.

Problem 1: Convert $(110101.101010)_2$ to Octal.

Answer: $(65.52)_8$

Problem 2: Convert $(10101111001.0111)_2$ to Octal.

Answer : $(2571.34)_8$

Conversion of Number System



Octal to Decimal Conversion

To convert a given octal number to a decimal number, multiply each digit in the octal number by the weight of its position and add all the product terms.

Problem 1 : Convert $(4057.06)_8$ to Decimal

Answer: $(2095.0937)_2$

Problem 1: Convert $(5497)_{10}$ to Binary.

Answer: $(001010101111001)_2$

Successive division

8	5497
8	687
8	85
8	10
8	1
	0

Remainders

↑ 1
| 7
| 5
| 2
1

$$5497_{10} = 12571_8 = 001010101111001_2.$$

Conversion of Number System



Problem 1: Convert $(101111010001)_2$ to Decimal.

Answer: $(3025)_{10}$

Since the given binary number is large, we first convert this number to octal and then convert the octal number to decimal.

$$101111010001_2 = 5721_8 = 5 \times 8^3 + 7 \times 8^2 + 2 \times 8^1 + 1 \times 8^0 = 2560 + 448 + 16 + 1 = 3025_{10}$$

Binary to Hexadecimal Conversion

To convert a binary to hexadecimal number, starting from binary point, make group of 4 bits each, on either side of the binary point and replace each 4-bit group by the equivalent hexadecimal digits.

Problem 1 : Convert $(1011011011)_2$ to Hexadecimal

Answer: $(2DB)_{16}$

Given binary number is

Groups of four bits are

Convert each group to hex

The result is

1011011011		
0010	1101	1011
2	D	B
$2DB_{16}$		

Conversion of Number System



Problem 1 : Convert $(01011111011.011111)_2$ to Hexadecimal

Answer: $(2FB.7C)_{16}$

Given binary number is

Groups of four bits are

Convert each group to hex

The result is

01011111011.011111					
0010	1111	1011	.	0111	1100
2	F	B	.	7	C
$2FB.7C_{16}$					

Hexadecimal to Binary Conversion

To convert a hexadecimal number to binary, replace each hex digit by its 4-binary group.

Problem 1 : Convert $(4BAC)_{16}$ to Binary

Answer: $(0100101110101100)_2$

Given hex number is

Convert each hex digit to 4-bit binary

The result is

4	B	A	C
0100	1011	1010	1100
0100101110101100_2			

Conversion of Number System



Hexadecimal to Binary Conversion

Problem 1 : Convert $(3A9E.B0D)_{16}$ to Binary

Answer: $(0011101010011110.101100001101)_2$

Given hex number is

Convert each hex digit to 4-bit binary

The result is

3	A	9	E	.	B	0	D
0011	1010	1001	1110	.	1011	0000	1101
$0011101010011110.101100001101_2$							

Hexadecimal to Decimal Conversion

To convert a hexadecimal number to decimal, multiply each bit in the hex number by its position weight and add all those product terms.

Problem 1 : Convert $(5C7)_{16}$ to Decimal

Answer: $(1479)_{10}$

Multiply each digit of 5C7 by its position weight and add the product terms.

$$\begin{aligned} 5C7_{16} &= (5 \times 16^2) + (12 \times 16^1) + (7 \times 16^0) \\ &= 1280 + 192 + 7 \\ &= 1479_{10} \end{aligned}$$

Conversion of Number System



Hexadecimal to Decimal Conversion

Problem 1 : Convert $(A0F9.0EB)_{16}$ to Decimal

Answer: $(41209.0572)_{10}$

$$\begin{aligned} A0F9.0EB_{16} &= (10 \times 16^3) + (0 \times 16^2) + (15 \times 16^1) \\ &\quad + (9 \times 16^0) + (0 \times 16^{-1}) + (14 \times 16^{-2}) + (11 \times 16^{-3}) \\ &= 40960 + 0 + 240 + 9 + 0 + 0.0546 + 0.0026 \\ &= 41209.0572_{10} \end{aligned}$$

Problem 2: Convert $(49056)_{10}$ to Binary

Answer: $(1011111110100000)_2$

Successive division

16	49056
16	3066
16	191
16	11
	0

Remainder

<i>Decimal</i>	<i>Hex</i>	<i>Binary group</i>
0	↑ 0	0 0 0 0
10	A	1 0 1 0
15	F	1 1 1 1
11	B	1 0 1 1

Therefore, $49056_{10} = BFA0_{16} = 1011,1111,1010,0000_2$.